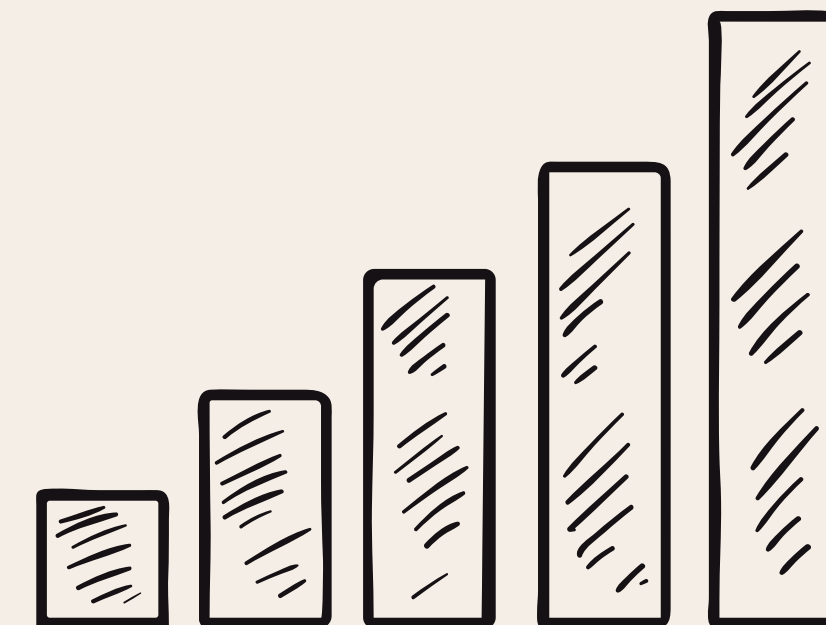
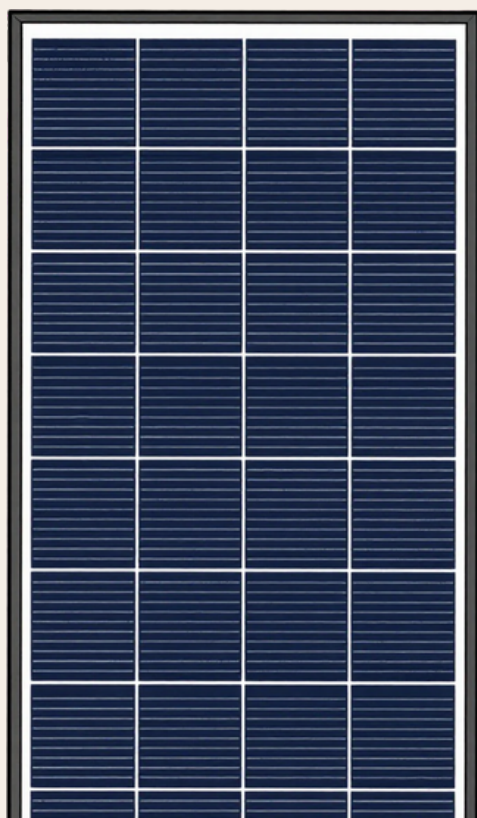
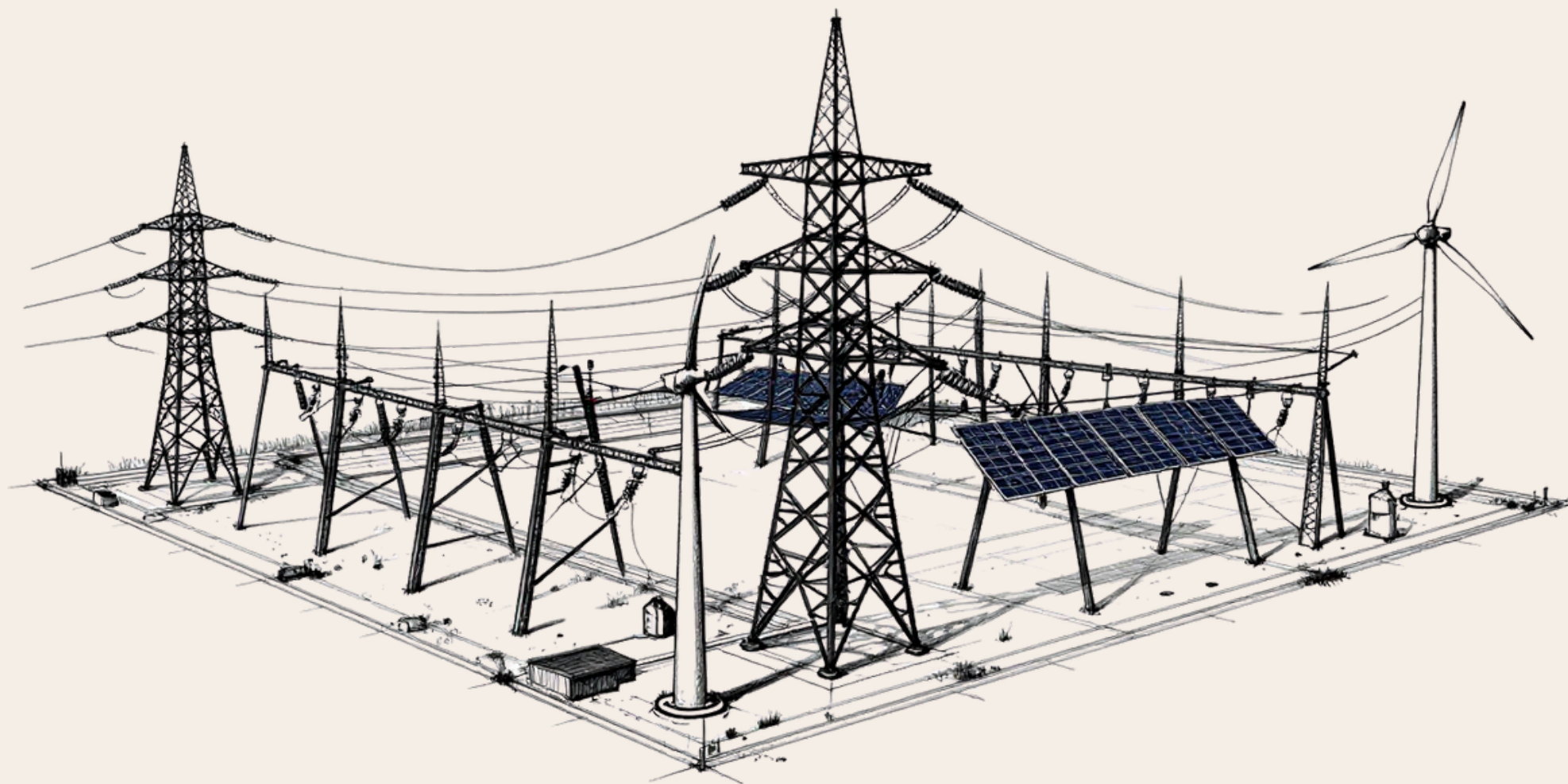
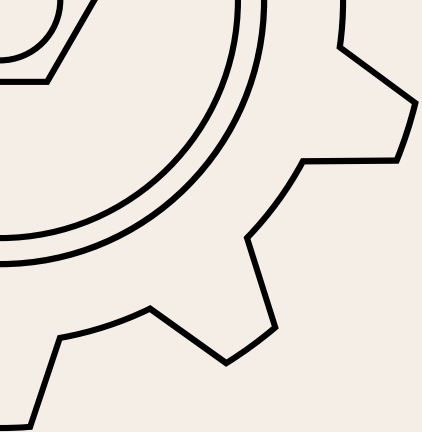




GRIDVISION AI – INTELLIGENT GRID LOAD OPTIMISATION & RENEWABLE ENERGY PERFORMANCE ADVISOR

An Agentic Solution for Clean Energy Curtailment Minimization and Asset Reliability





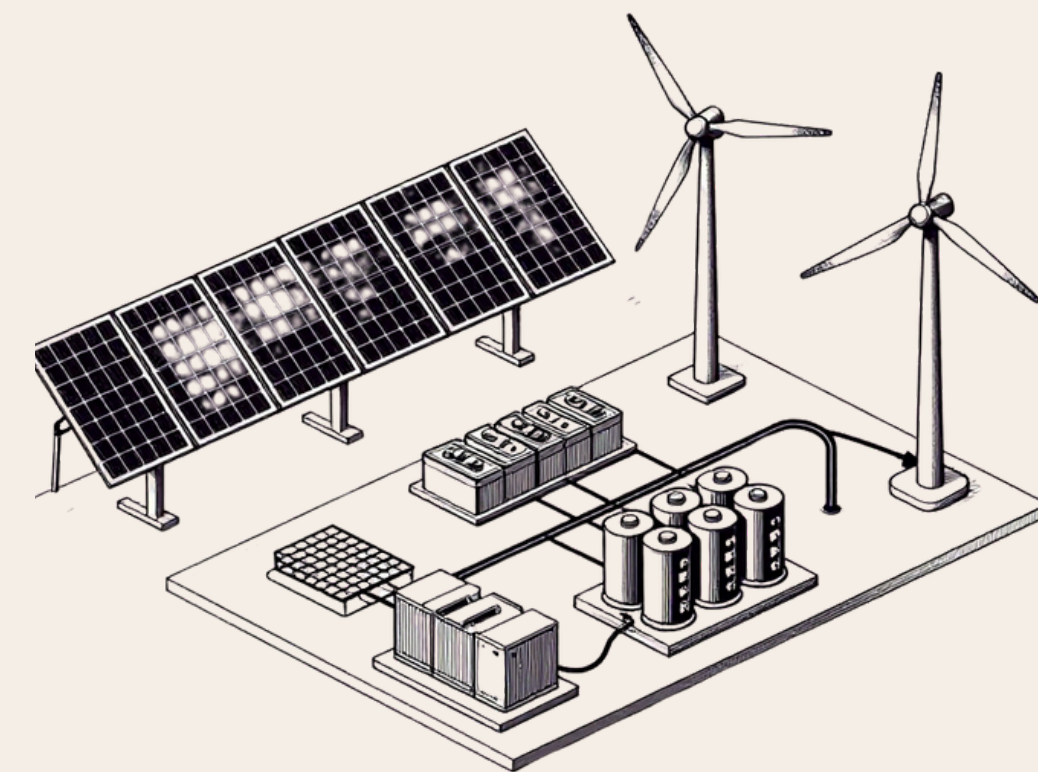
Track: *Utilities — Problem Statement U2: Grid Load
Optimisation & Renewable Energy*

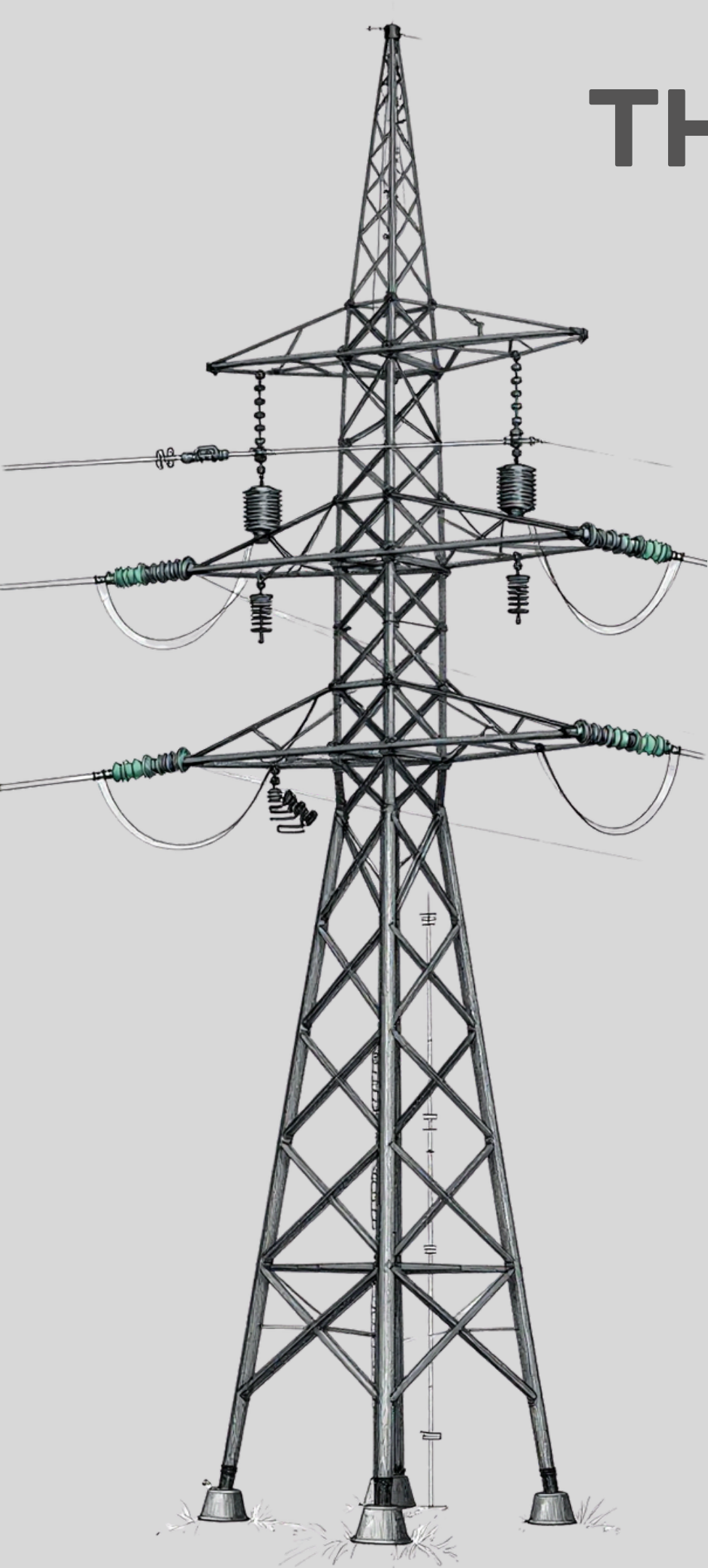
Performance Advisor

Team Name: *GridVision-AI*

Repository: *<https://github.com/Aryan-223/bob-ai-hackathon-Team>*

Event: *IBM BoB AI Innovation Hackathon 2026*





THE CLEAN ENERGY PARADOX: GRID INSTABILITY & WASTED POWER

- ***The Core Problem:***

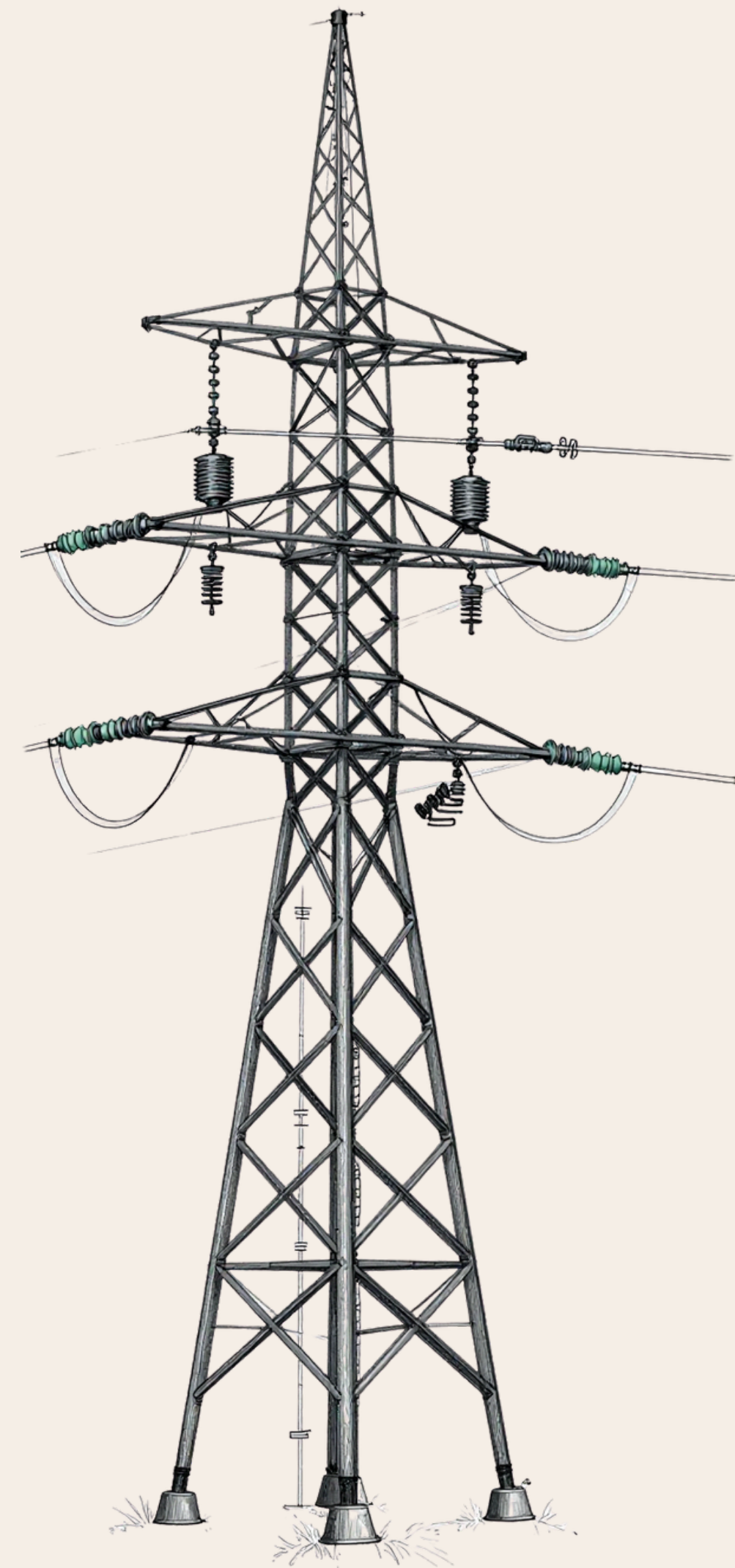
Modern power systems face severe net-load volatility ("The Duck Curve"). Midday solar peaks saturate regional transmission lines, while steep evening demand ramps strain grid frequency regulation.

- ***Who Has This Problem?***

Transmission System Operators (TSOs), Regional Distribution Utilities, and Renewable Farm Asset Managers.

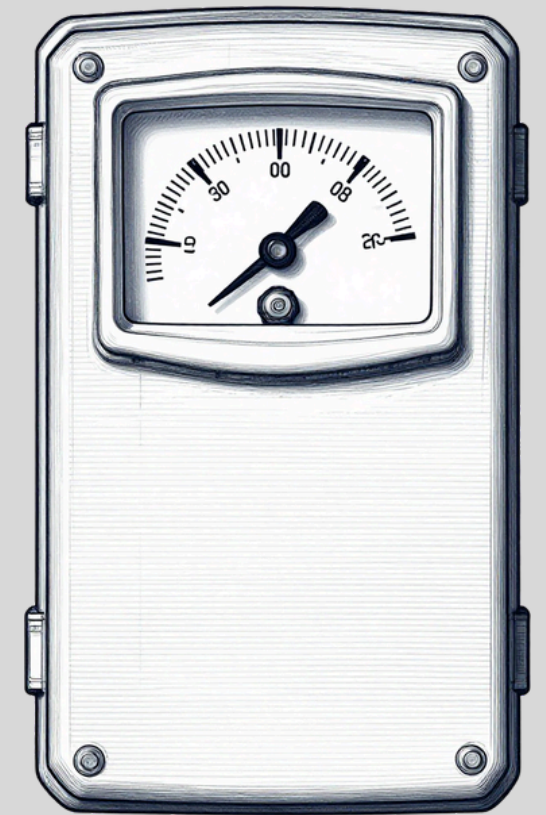
• *Why It Matters?*

- Massive Clean Energy Waste: Over ~8 TWh of zero-carbon clean energy was curtailed (deliberately switched off) in the US alone in 2023 because the grid could not absorb it.
- High Outage & Peaker Plant Costs: Sudden evening demand spikes require expensive, high-emission fossil peaker plants to ramp rapidly, driving grid instability.
- Blind Asset Operations: Solar and wind generation anomalies (such as inverter thermal clipping or blade pitch misalignment) go unnoticed in real time, degrading asset yield by 10–25%.



Key Solution Pillars:

- 1. *Dual-Horizon Forecasting:*** Anticipates net load ramps and clean generation surpluses 24 hours in advance.
- 2. *Curtailment Reduction Engine:*** Schedules Battery Energy Storage Systems (BESS) to charge during midday overgeneration and discharge during evening peaks.
- 3. *Asset Diagnostics:*** Continuously monitors telemetry against theoretical physical baselines to catch faulty inverters or soiling before failure.
- 4. *Operator Briefing Agent:*** Automatically produces natural-language Bottom-Line-Up-Front (BLUF) operational memos.



Introducing GridVision AI — Autonomous Decision Support for Grid Operators

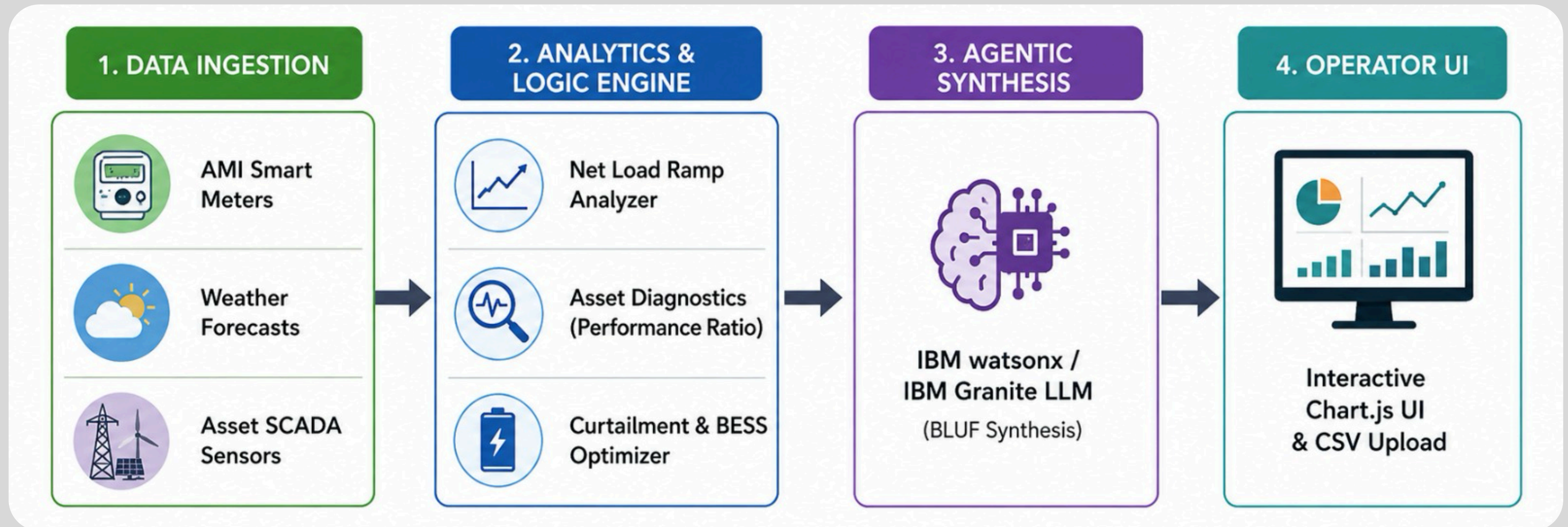
Core Value Proposition:

An AI-powered co-pilot that simultaneously forecasts demand spikes, eliminates curtailment through automated storage dispatch, flags underperforming assets with root-cause diagnoses, and synthesizes shift-ready briefs for control rooms.

Key Solution Pillars:

1. Dual-Horizon Forecasting: Anticipates net load ramps and clean generation surpluses 24 hours in advance.
2. Curtailment Reduction Engine: Schedules Battery Energy Storage Systems (BESS) to charge during midday overgeneration and discharge during evening peaks.
3. Asset Diagnostics: Continuously monitors telemetry against theoretical physical baselines to catch faulty inverters or soiling before failure.
4. Operator Briefing Agent: Automatically produces natural-language Bottom-Line-UpFront (BLUF) operational memos.

END-TO-END TECHNICAL ARCHITECTURE & AGENTIC FLOW



KEY INSIGHTS

- ***Ingestion Layer:*** Ingests regional hourly demand, solar irradiance (GHI), wind speeds, and inverter temperatures.
- ***Computational Core [src/analytics.py]:*** Calculates net demand, evaluates the Performance Ratio ($PR = P_{\text{actual}} / P_{\text{theoretical}}$), and solves the storage dispatch equations.
- ***Backend API [src/app.py]:*** Lightweight Flask service providing structured REST endpoints (/api/forecast, /api/anomalies, /api/dispatch).
- ***LLM Synthesis Engine:*** Translates raw numerical data into actionable dispatch briefs using structured Granite prompt templates.



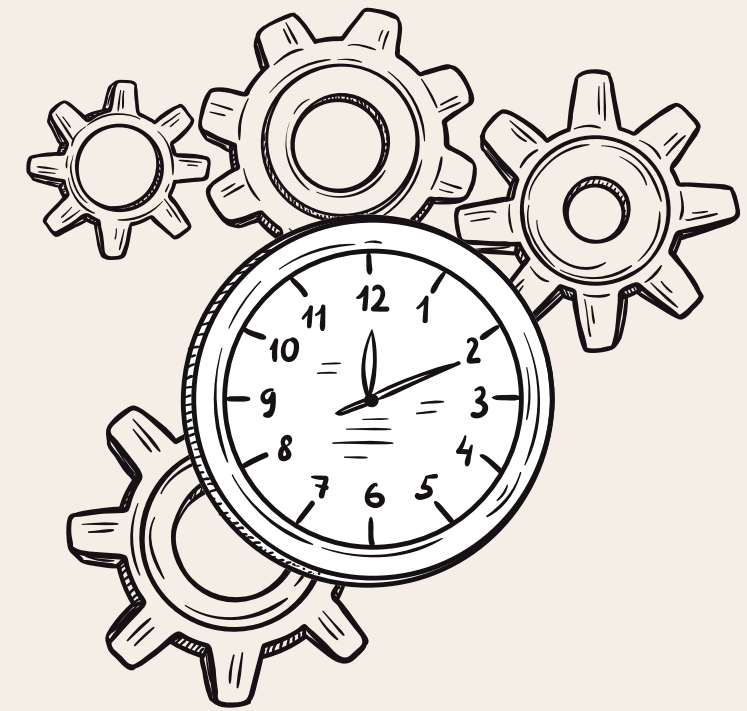
DEEP DIVE: ANOMALY DIAGNOSTICS & CURTAILMENT MITIGATION

Scenario Walkthrough (Real-Time Control Room Simulation):

- 1.Spike & Curtailment Detection:** The engine detects a 48 MW net-load ramp between 17:00 and 19:00, alongside a 35 MW midday renewable surplus at 13:00.
- 2.Autonomous Mitigation:** BESS schedules a 15 MW charge cycle during the midday window (absorbing 42 MWh of clean power that would otherwise be curtailed) and schedules a discharge cycle during the evening spike.
- 3. Asset Anomaly Flagged:** SCADA sensor readings at Solar Array 3 trigger an alert: Irradiance is 980 W/m², but output is pinned at 12 MW with inverter temperature at 78.5°C.
- 4. Root Cause Diagnostic:** System classifies the event as Thermal Derating / Heat Exchanger Clogging, alerting operators to dispatch targeted maintenance.

IBM TECHNOLOGIES INTEGRATION

Built with IBM Bob & IBM watsonx



- ***IBM Bob IDE (Required Core Component):***
 - Served as the pair programmer across the full software development lifecycle.
 - Used Plan Mode to architect system modules and Code/Agent Mode to implement data ingestion, math formulations, Flask endpoints, and test suites.
 - Fully tracked in compliance with hackathon rules: exported task session logs and token/cost summary screenshots are stored in bob_sessions/.
- ***IBM watsonx.ai & Granite Models:***
 - Utilized granite-3-8b-instruct within the Prompt Lab to build the dispatch reasoning pipeline.
 - Generates concise, structured operational summaries directly from intermediate JSON outputs, avoiding hallucination and minimizing compute usage.
- ***Project Governance:***
 - Implemented strict credential safety via local .env handling to prevent cloud API key exposure.

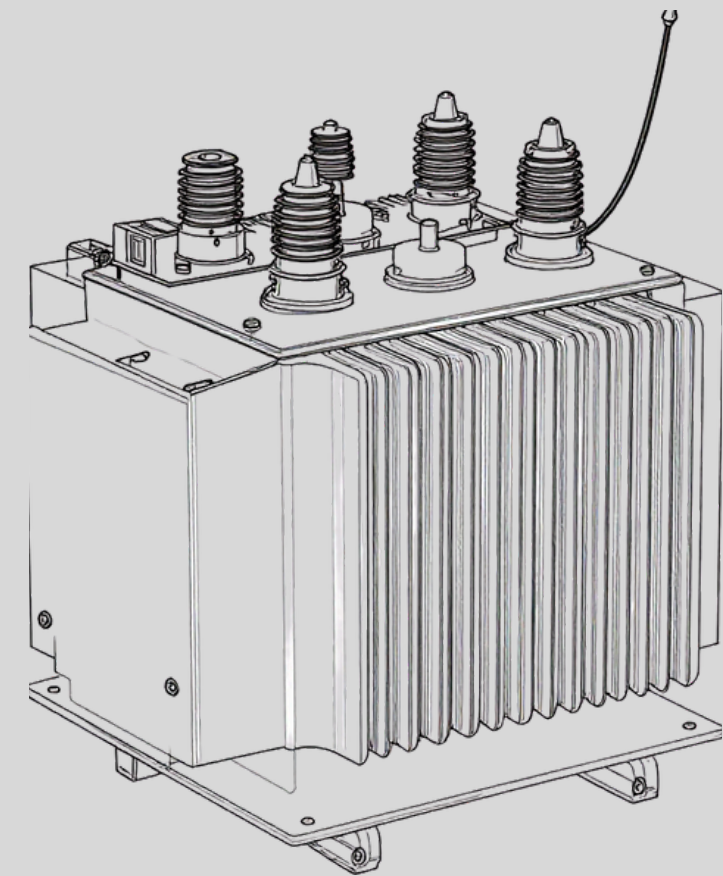
QUANTIFIABLE BUSINESS VALUE & GRID OUTCOMES

Target Metrics:

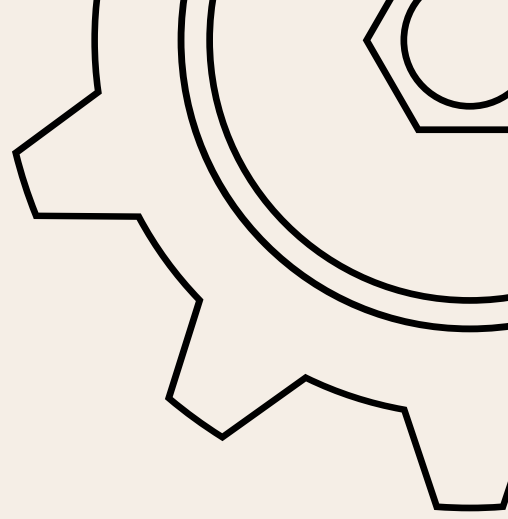
- Curtailment Avoidance: Up to 70–85% reduction in wasted local renewable energy via automated battery dispatch scheduling.
- Peak Ramp Mitigation: Lowers peak ramp reliance on fossil peakers by 15–20%, stabilizing grid frequency.
- Rapid Root-Cause Identification: Cuts asset anomaly triage time from days to seconds using real-time telemetry comparison.
- Faster Decision-Making: Delivers structured shift briefs to operators in under 3 seconds, reducing manual spreadsheet cross-referencing.

Future Roadmap:

- Multi-agent grid coordination via IBM watsonx Orchestrate ADK.
- Integration with live regional SCADA feeds and automated demand-response market signals.



Team Members & Project Roles



Aryan Jariwala(25EE015): Primary Developer & Bob Lead: Guided IBM Bob IDE, designed backend analytics in src/analytics.py, managed Bob session tracking in bob_sessions/.

Meet Patel(25EE038): AI Prompt Engineer & Data Modeler: Formulate mathematical specifications in docs/specifications.md, structured synthetic telemetry datasets, refined IBM Granite prompt pipelines.

Shreya Kumari(25EE020): Frontend & UI/UX Designer: Developed Chart.js dashboard layout, styled alert components, and built the CSV upload interface.

Keyushi Patel(25EE036): Technical Writer & Presenter: Structured README.md documentation, authored slide deck, and recorded final demo walkthrough.